NATIONAL ECONOMICS UNIVERSITY
FACULTY OF DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

# GROUP ASSIGNMENT REPORT

Midterm

## Web Design and Programming (CNTT1164)

# Rùa Toán Basic: A Mathematics Practice Website

Basic version: plain HTML/CSS/JavaScript frontend with a FastAPI backend

**Group:** Group 09
**Class:** EP16.CNTT1164(126)_01
**Instructor:** M.Sc. Le Duy Khanh
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**GitHub:** https://github.com/thunopro/ruatoan_basic
**Demo:** https://ruatoanbasic.vercel.app

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## Declaration

We declare that this report is the result of our group's own work for the course Web Design and Programming. All sources (literature, source code, images, and AI tools) are cited explicitly. We take full responsibility for the submitted content.

## Task Allocation and Contributions

| Member | Work done | Contribution (%) |
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| Nguyen Khac Thu | Full-stack architecture, FastAPI backend, PostgreSQL schema, plain HTML/CSS/JS frontend, deployment, smoke testing, and comprehensive report writing | 100 |
| Le Quang | Team member (no implementation contribution in basic deliverable) | 0 |
| Huy Minh | Team member (no implementation contribution in basic deliverable) | 0 |
| | **Total** | **100** |

*Contribution percentages sum to 100% as required by the course guidelines.*

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# 1. Introduction

## 1.1. Project overview

Rùa Toán Basic is a mathematics practice website for lower secondary students. It is the "basic" version of the Rùa Toán product: the same educational goal and the same content catalogue, but rebuilt with a much simpler and more transparent technology stack. The frontend is plain HTML, CSS, and JavaScript with no framework and no build step; the backend is a small FastAPI application running on Vercel's Python runtime; user accounts and uploaded photographs are stored in PostgreSQL. The content catalogue (exam collections, problems, lessons, books, and documents) is still read from versioned JSON files, exactly as in the original project, so the two versions share one source of truth for learning material.

The live demo of this basic version is published at `https://ruatoanbasic.vercel.app`. The original, more feature-rich version remains at `https://ruatoan.vercel.app` and is kept unchanged; the basic version does not replace it. Instead, the basic version exists to make every layer of the application inspectable and maintainable: a student or instructor can open the browser's developer tools and read the network requests, the DOM, and the CSS directly, and can read the entire backend in five short Python files.

## 1.2. Motivation

The original Rùa Toán application is built with Next.js, React, TypeScript, and Auth.js. That stack is powerful, but it hides a great deal of behaviour behind a framework and a build pipeline: server components, client components, route handlers, a content cache, and an OAuth session layer. For a course on web design and programming, and for a team that wants to understand exactly how a request travels from the browser to the database and back, that opacity is a genuine cost. The basic version removes it.

Three concrete motivations drove the rebuild:

1. **Inspectability.** Every page is a single HTML file, every interaction is a short JavaScript function, and every backend route is a short FastAPI handler. A reader can follow a feature from its button to its SQL statement without crossing framework boundaries.
2. **Deployment clarity.** Vercel supports FastAPI through its Python runtime, but the routing, bundling, and cold-start behaviour differ from the Node.js runtime. Rebuilding against that runtime forced an explicit, documented `vercel.json` and a small, auditable set of dependencies.
3. **A controlled scope.** The original application contains advanced features (Google OAuth, a device-token photo API, a content crawl and preparation pipeline) that are not all needed for the core learning loop. The basic version keeps the learning loop—browse collections, read problems, study lessons, consult the library, mark problems, submit photographs—and drops the rest, while recording those omissions honestly in Chapter 4 (Section 4.4).

## 1.3. Objectives

The basic version was built to satisfy the following objectives:

O1. Provide the core learning loop: browse exam collections, read individual problems with mathematical notation, study lessons, and consult books and documents.

O2. Support filtering and search: by grade, topic, collection kind, year, province, and free text, with pagination.

O3. Render mathematics correctly in the browser using MathJax, including in problem statements and lessons.

O4. Allow a learner to register, log in, and log out with an email and password, with the session held in a signed cookie.

O5. Allow a logged-in learner to upload, view, and delete photographs of written work, with the images stored in PostgreSQL.

O6. Let a learner mark problems in the browser and keep that list locally, without requiring an account.

O7. Deploy successfully on Vercel, with durable storage in PostgreSQL rather than in the function's temporary filesystem.

O8. Keep the source small, readable, and free of a build step.

## 1.4. Scope and boundaries

Implemented in this version:

- Home page with live catalogue totals and a list of marked problems.
- Practice page: text/grade/topic filters, 20 results per page, problem cards, and a "mark / unmark" toggle.
- Exam collections page: paginated list (20 per page) and a collection detail page showing problems in their original order, with source attribution and a link to the source.
- Lessons page: client-side search over 90 lessons, expandable bodies.
- Library page: books (216) and documents (89) with client-side search and external links.
- Account pages: registration and login with email and password; logout.
- Submission page: upload up to six images (1.5 MB each) per request, view them, and delete them.
- Catalogue of 3,381 collections and 33,929 problems, unchanged from the original.

Explicitly *not* implemented in this version, by design:

- Google OAuth / social sign-in (the original has it; this version uses email and password only).
- Device-token based pseudonymous photo ownership (this version ties photos to the logged-in account).
- Cross-device synchronisation of marked problems (they stay in `localStorage`).
- Automatic answer marking, a teacher administration panel, or a learning analytics dashboard.
- The content crawl and preparation pipeline (the JSON is reused as-is).

## 1.5. Technology overview

The stack is deliberately small. Table 1 lists each layer, the technology, and its responsibility in Rùa Toán Basic.

## 1.6. Catalogue size

The content catalogue is reused unchanged from the original project. The inspected JSON files contain 3,381 collection records and 33,929 nested problem records, together with 90 lessons, 216 book, magazine, or resource records, and 89 document records. That is 37,705 raw records when each collection, problem, lesson, book, and

| Layer | Technology | Use in Rùa Toán Basic |
|---|---|---|
| Markup | Plain HTML5 | Eight pages (home, practice, exams, lesson, library, login, register, submissions); semantic markup, no template engine |
| Styling | Plain CSS3 | Unified styling: `public/css/original.css` (core visual layout, branding, cards, footer) + `public/css/style.css` (design tokens, forms, photo grid, a11y, dark mode) |
| Behaviour | Plain JavaScript (ES2020) | Eight small files (`public/js/*.js`); modular, no framework, no bundler, no build step |
| Mathematics | MathJax 3 (CDN) | Client-side LaTeX rendering of inline `$...$` and display `$$...$$` |
| Backend | FastAPI + Uvicorn | `backend/main.py` and helpers `auth.py`, `content.py`, `database.py`; about 600 clean lines |
| Database | PostgreSQL via `psycopg` | Schema `ruatoan_basic`: tables `users`, `photos`, and `password_resets` |
| Catalogue | JSON files | 3,381 collections, 33,929 problems, 90 lessons, 216 books/resources, 89 documents |
| Hosting | Vercel (Python runtime) | Root `main.py` entry point, Vercel function routing, automated CI/CD |

**Table 1.** Technologies used in Rùa Toán Basic.

document entry is counted once. The JSON files occupy roughly 16 MB on disk (after removing a 19 MB duplicate copy, see Section 3.8). A problem may carry more than one topic tag, so topic counts are assignments and do not partition the problem set.

## 1.7. Report evidence and snapshot

This report is based on a source-code audit and live verification of the Rùa Toán Basic deployment on 1 November 2026, following architectural unification, security hardening, and accessibility fixes. The following checks were run and verified:

- `node --check` on all eight JavaScript files (all passed).
- `python3 -m py_compile` on all Python backend and script files (all passed).
- Comprehensive automated smoke test: `/`, `/api/stats`, `/api/daily`, `/api/collections`, `/api/problems`, `/api/lessons`, `/api/books`, and `/api/docs` all returned HTTP 200 with valid payloads.
- Complete account and security flow: user registration, credential validation, session cookie issuance, authenticated `/api/me`, password change, database-persisted password reset via single-use tokens, and logout all executed end to end.
- Photo submission pipeline: magic-byte validation (PNG/JPEG/WebP), size bounds (1.5 MB), database binary storage, retrieval with `nosniff`, and ownership-verified deletion.
- Browser runtime audit: zero console errors across all eight pages in production, zero 404 asset failures (fonts and favicons fully resolved), and validated keyboard accessibility (`:focus-visible`) and high-contrast inputs.

Database credentials and session secrets are managed strictly through environment variables (`.env.local` and Vercel environment settings) and never committed to source control.

# 2. Analysis and Design

## 2.1. Requirements

The requirements below describe visible behaviour and quality goals represented in the inspected source. They are not claims that every case has passed a formal acceptance test beyond the smoke test described in Chapter 4.

### 2.1.1. Functional requirements

FR1. Browse exam collections with pagination (20 per page) and open a collection to read its problems in original order, with source attribution and a link to the source.

FR2. Browse individual problems by free text, grade, and topic; open a problem with its collection context.

FR3. Read problem and lesson statements containing inline or display mathematical notation rendered with MathJax.

FR4. Browse the five topic areas' lessons (90), and the library of books (216) and documents (89), with client-side search and external links.

FR5. Register, log in, and log out with an email and password; keep the session in a signed cookie.

FR6. Upload up to six photographs (1.5 MB each) per request, view them, and delete them, for a logged-in user.

FR7. Keep a compact list of marked problems in browser storage, independently of the account and the database.

FR8. Show live catalogue totals on the home page.

### 2.1.2. Non-functional requirements and constraints

- **Responsive presentation:** pages must remain readable on narrow and wide screens. The CSS contains responsive layout rules; a formal device test has not been recorded.
- **Readable mathematics:** problem text is rendered with MathJax. No print benchmark was collected (the basic version has no dedicated print view).
- **Source transparency:** collection pages show source names and link to source pages when URLs are available.
- **Keyboard and assistive-technology support:** the layout includes visible keyboard focus (`:focus-visible`), labelled navigation and controls, and `aria-label` on inputs without a visible label. These are implementation details, not evidence of a completed accessibility audit.
- **Bounded image input:** the API accepts PNG/JPEG/WebP only, no more than six files per request, 1.5 MB per file, and 50 images per account in total.
- **Account-scoped photo ownership:** photos belong to the logged-in user. Clearing the browser does not lose them; they are stored in PostgreSQL.
- **Secret handling:** database and session secrets are supplied through environment variables and must not be committed or printed in the report.
- **Content quality:** records preserve source context, but the project has no measured editorial audit that proves every transcription, link, tag, or year is correct.

## 2.2. Interface design

The interface is structured around the authentic Rùa Toán design system: a deep forest green background (`#0F3D2E`, `--bg`), dark panels (`#164A38` / `#163523`), and high-contrast warm parchment paper cards (`#FFF8E7`, `--surface`)

for readable mathematical notation. The visual identity highlights lively turtle-themed accents: vibrant lime (`#B6FF3B` / `#A3E635`, `--ooze`), warm gold (`#FBB03B`), and terracotta coral (`#FF6F59`). Typography pairs Google Font **Bungee** for playful, retro gaming-style headers, collection badges, and numeric stats with **Be Vietnam Pro** for clean, highly legible Vietnamese content and mathematical explanations.

Every page is ported to a uniform and modular layout structure:

- `top/header`: Sticky header containing the brand logo, navigation links (*Luyện tập*, *Đề thi*, *Bài học*, *Thư viện*, *Nộp bài*), and session state indicator ("Đăng nhập" or "Đăng xuất (Tên)").
- `phead`: High-impact hero section presenting page title, subheadings, and quick action buttons.
- `stats` and `mission`: Visual metric counters and orientation banners for learners.
- `topics`: Pill-based interactive filter controls for grade levels and mathematical domains.
- `grid/ccard`: Neomorphic, tactile cards for collections and problem statements with 3D drop-shadow borders (`3px solid var(--ink)`).
- `pager`: Paginated navigation controls with "Xem thêm" infinite-append or index jump.
- `foot`: Grounded footer with credits, license, and navigation links.

The stylesheet automatically adapts to `prefers-color-scheme: dark` via CSS custom properties, with hardened input contrast, keyboard accessibility outlines (`:focus-visible`), and light-on-dark/dark-on-light readability overrides.

The figures below are screenshots captured directly from the live public deployment on 1 November 2026 (desktop viewport at 1440 px, and mobile viewport at 390 px). They exhibit the authentic green/parchment visual theme, responsive layout, interactive filter pills, and mathematical typesetting. Specifically, Figure 1 presents the primary home dashboard; Figure 2 displays the problem practice feed; Figure 3 showcases the curated exam collection catalogue; Figure 4 illustrates the searchable theory lessons; Figure 5 highlights the reference mathematics library; Figure 6 shows the student homework photo submission portal; Figure 7 shows the secure authentication view; and Figure 8 verifies full mobile responsiveness at 390 px viewport width.




**Figure 1.** Public demo home page: primary actions, catalogue totals, and recent collections.

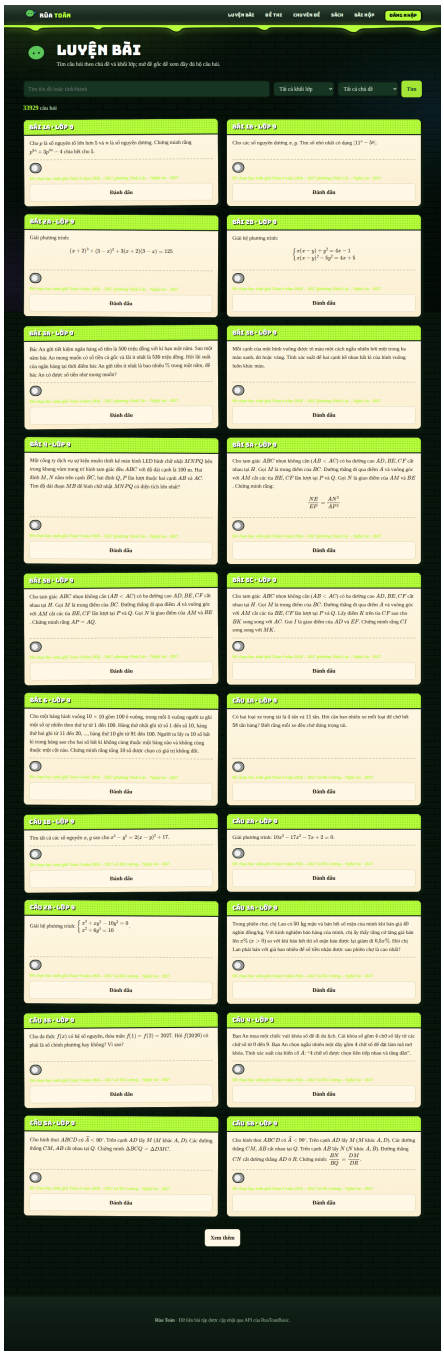



**Figure 2.** Practice page at desktop width, showing topic navigation and problem cards.




**Figure 3.** Exam collection page with ordered statements and source attribution.




**Figure 4.** Lesson page with client-side search and expandable bodies.

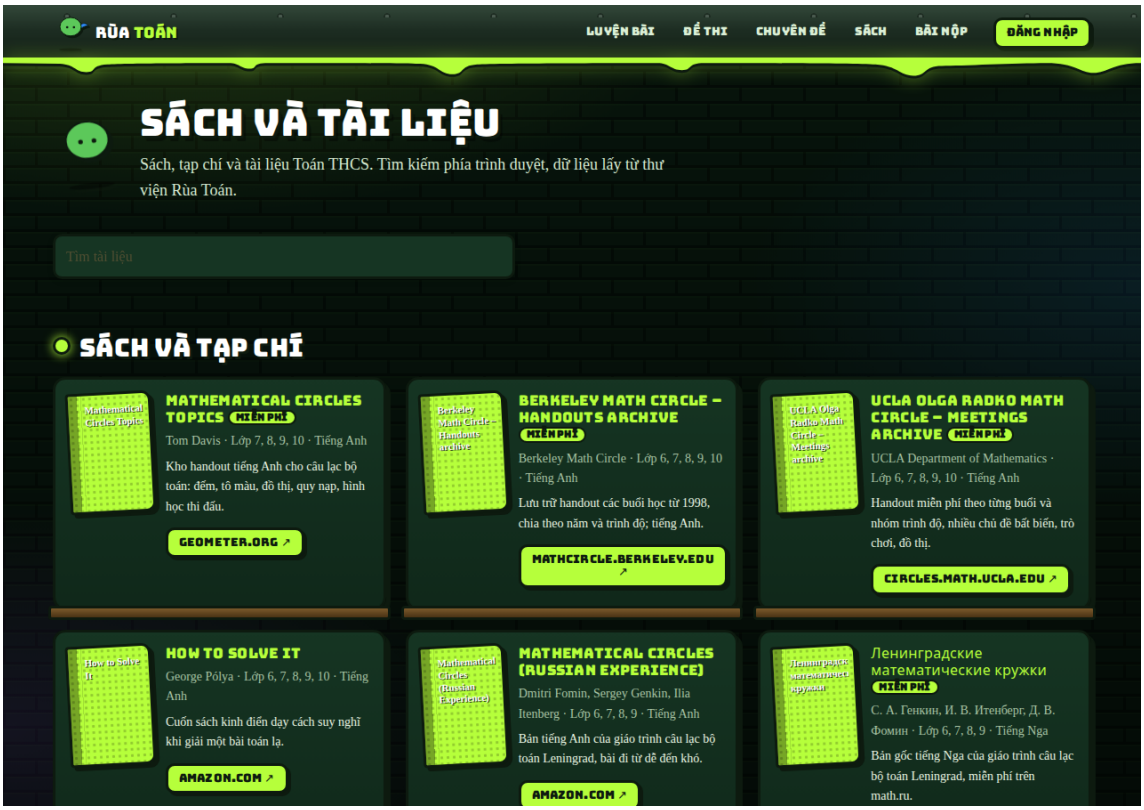



**Figure 5.** Library page combining books and documents with client-side search.

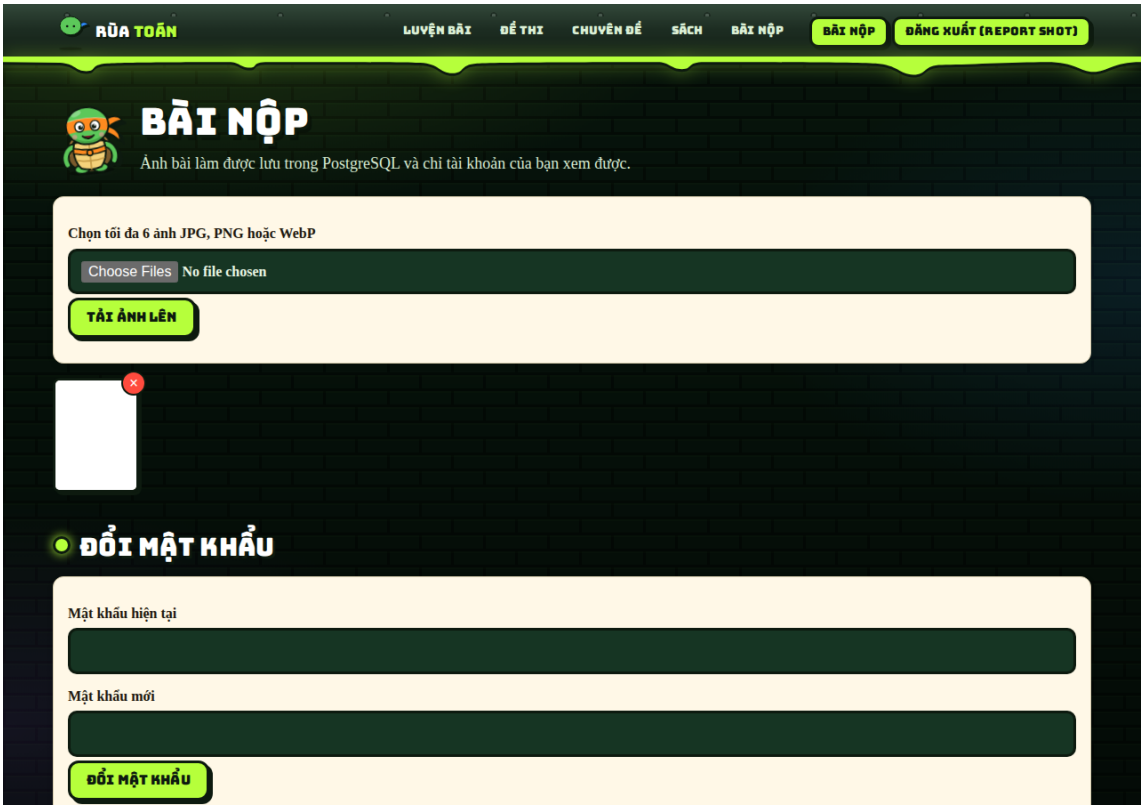



**Figure 6.** Submission page for uploading, viewing, and deleting photographs.

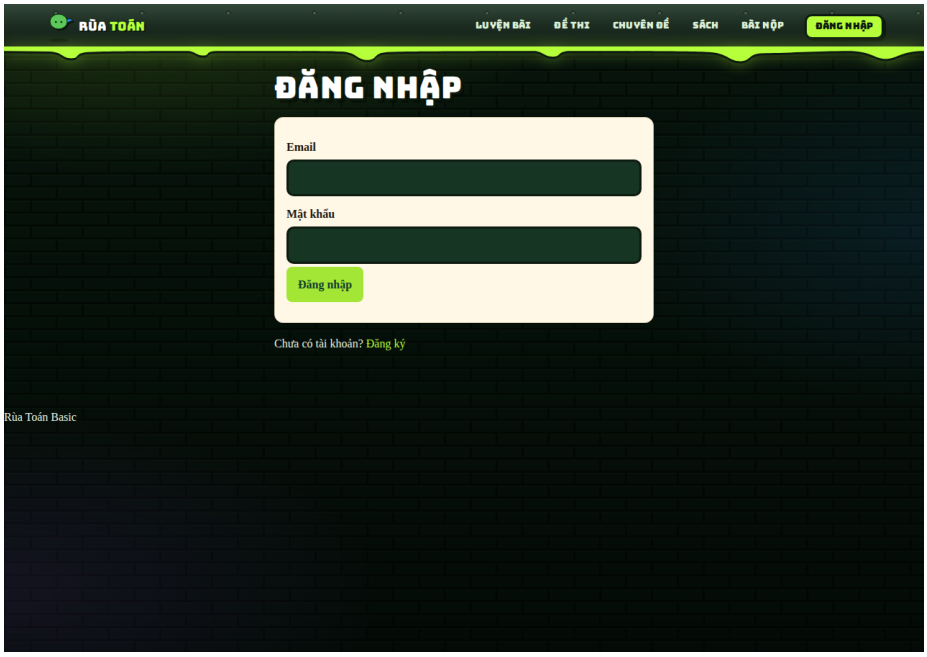



**Figure 7.** Login page with email and password.




**Figure 8.** Home page captured at a 390-pixel viewport width.

## 2.3. System architecture

Figure 9 summarises the system boundary of the basic version. The browser requests static files and JSON-backed pages from Vercel; Vercel serves the static assets from `public/` and routes `/api/*` to the FastAPI function. FastAPI reads the catalogue JSON from disk (cached in memory per process) and the user data from PostgreSQL. There is no OAuth provider and no build step.

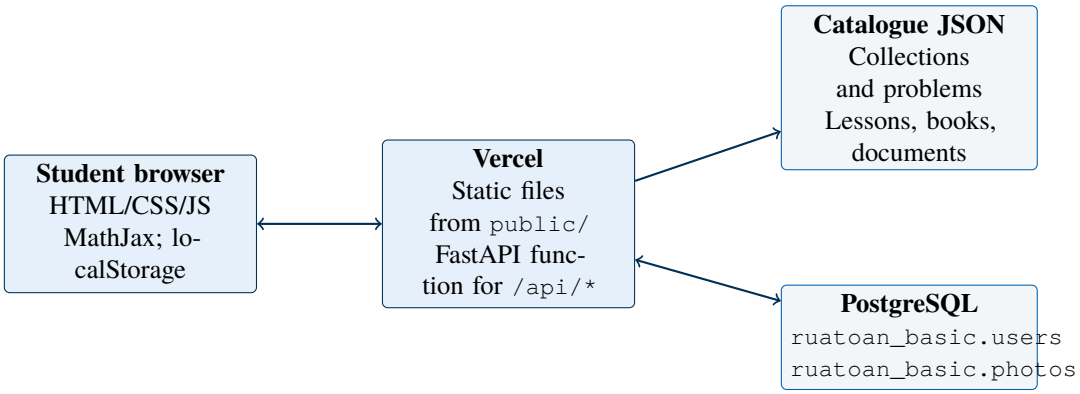



**Figure 9.** Logical system architecture of Rùa Toán Basic. Content, account data, and photos use separate paths.

The FastAPI application declares no revalidation interval (it is not a cached framework page); instead, the catalogue is held in an in-memory cache keyed by file modification time, so a redeploy refreshes it automatically. The content module also folds Vietnamese diacritics once per problem at load time, so search is fast per request.

## 2.4. Data design

The catalogue is organised as structured JSON, exactly as in the original project. The groups and primary fields are summarised in Table 2. There are eleven non-sample collection files under `data/sources/`, seven non-sample lesson files under `data/theory/`, and top-level files `books.json` and `docs.json`. In this snapshot these JSON files occupy about 16 MB (after a 19 MB duplicate copy was removed, see Section 3.8). Files beginning with an underscore are samples or work-in-progress and are skipped by the loader.

| Record group | Count | Primary fields / use |
|---|---|---|
| Collections | 3,381 | ID, kind, title, grade, optional year/province, source name/URL, ordered problems |
| Problems | 33,929 | ID, label, statement text, topic tags; nested in a collection |
| Lessons | 90 | ID, topic, title, summary, body, optional examples |
| Books and resources | 216 | ID, title, author/publisher, grades/topics, description, URL, kind and source |
| Documents | 89 | ID, topic, title, author, grade/page metadata, description and source links |

**Table 2.** Catalogue groups counted directly from non-sample JSON records.

Figure 10 shows how the catalogue, browser state, photo rows, and user rows relate. The main nested relationship is *Collection contains an ordered array of Problems*. Collection metadata supports exam filtering and attribution. A problem carries its own stable identifier and display label, statement, and one or more topic tags. Books, lessons, and documents are separate runtime collections. The data model is intentionally source-aware: the collection retains source URL and source name so individual problem pages can display the origin.

The corpus spans five collection kinds: 1,195 HSG, 501 grade-10 specialised entrance, 733 Olympiad, 844 magazine, and 108 topic collections. The current grade metadata contains 237 grade-6, 342 grade-7, 411 grade-8, 1,786 grade-9, and 605 grade-10 collections. The recorded years range from 1935 to 2027. The collection-kind and grade distributions appear in Figures 11 and 12.

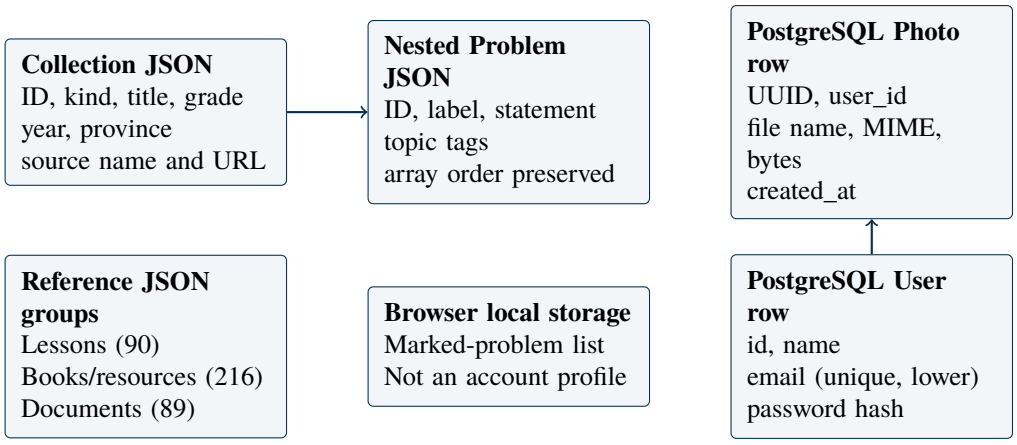



**Figure 10.** Catalogue and user-related records by storage location. Marked problems stay in the browser.

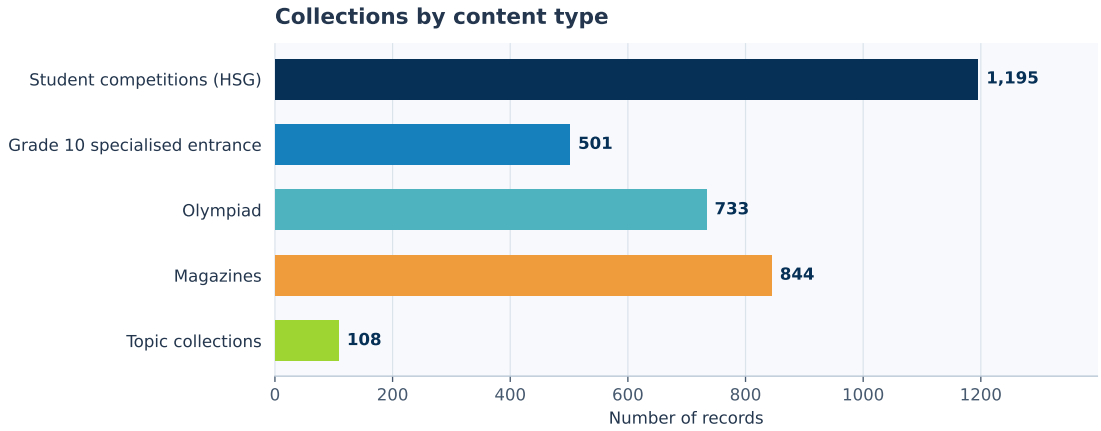



**Figure 11.** Collection counts by source kind in the inspected JSON corpus.

Topic tags are multi-valued. The current records contain 7,643 number-theory assignments, 12,677 algebra assignments, 3,926 inequality assignments, 10,442 geometry assignments, and 4,098 combinatorics assignments. Since a problem can carry more than one tag, these values are not a partition of the 33,929 problems. Figure 13 visualises these overlapping assignments.

## 2.5. Database design

PostgreSQL is used for mutable user accounts, password reset tokens, and photo submissions, rather than for the static catalogue. The application creates schema `ruatoan_basic` on first use and three tables:

- `users(id, name, email, password_hash, created_at)` with a unique index on `lower(email)` (case-insensitive uniqueness).
- `photos(id, user_id, file_name, content_type, byte_size, data, created_at)` with an index on `(user_id, created_at)` for per-user history ordered by time, cascading on user deletion.
- `password_resets(id, user_id, token_hash, expires_at, created_at)` with a unique index on `token_hash` and cascade deletion on user removal, enforcing single-use, 30-minute expiration for secure account recovery.

Photos are stored as `bytea` (binary) columns, so the function's temporary filesystem is never used for durable data—a requirement on Vercel, where the filesystem of a serverless function is ephemeral. There is no foreign key

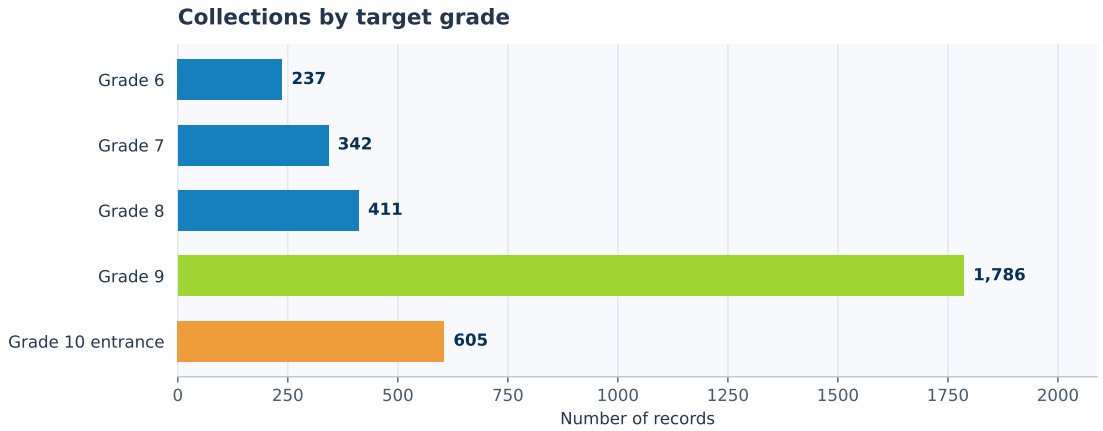



**Figure 12.** Collection counts by target grade; grade 10 denotes specialised entrance material.

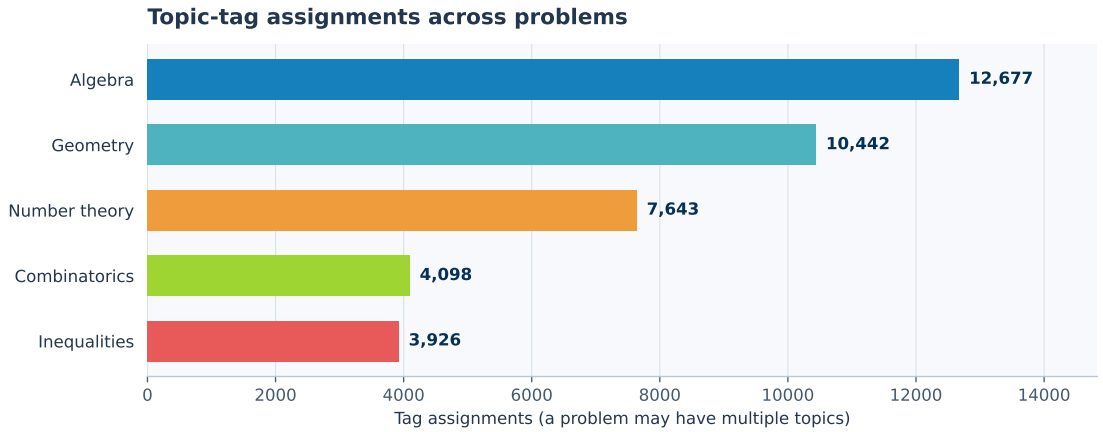



**Figure 13.** Topic-tag assignment counts. Problems with multiple tags contribute to multiple bars.

from `photos` to the JSON catalogue, because problems are not rows in PostgreSQL. The API does not verify that an uploaded problem ID exists in the catalogue; it stores whatever the client sends as metadata.

Passwords are hashed with PBKDF2-HMAC-SHA256 (200,000 iterations) using a 16-byte random salt, and stored in the format `pbkdf2_sha256$iterations$salt$hash` so the cost can be raised later without invalidating existing hashes. The session is a signed cookie (HMAC over a JSON payload containing the user id and an expiry), `HttpOnly`, `SameSite=Lax`, and `Secure` on HTTPS.

## 2.6. API design

The API is a small read/write surface. Table 3 lists every endpoint, its request shape, and its successful and error behaviour as confirmed by the source and by the smoke test.

For photo upload, the browser sends the raw selected files (no client-side compression in this version; the original project compressed them). The server validates the count (1–6), the per-file size (1.5 MB), and the magic bytes (PNG/JPEG/WebP) rather than trusting the declared MIME type. It enforces a per-account cap of 50 images, then inserts all validated files in one transaction. Accepted bytes are stored as `bytea`. The image endpoint sets `X-Content-Type-Options: nosniff` and returns the stored bytes.

| Method | Path | Request | Successful result / explicit errors |
|---|---|---|---|
| GET | `/api/stats` | none | `{collections,problems,lessons,books,docs,topics}` |
| GET | `/api/daily` | none | `{problem,collection}` for today's problem |
| GET | `/api/collections` | `?page,limit,grade,kind,year,province,search` | `{total,page,items}` (summary, no problem arrays) |
| GET | `/api/collections/{id}` | `?page,per` | collection metadata + one page of problems; 404 if missing |
| GET | `/api/problems` | `?page,limit,search,grade,topic` | `{total,page,items}` |
| GET | `/api/lessons` | `?search` | `{items}` |
| GET | `/api/books` | `?search` | `{items}` |
| GET | `/api/docs` | `?search` | `{items}` |
| POST | `/api/register` | JSON `{name,email,password}` | `{user}`; 400/409 on validation or duplicate email |
| POST | `/api/login` | JSON `{email,password}` | `{user}` + session cookie; 401 on bad credentials |
| POST | `/api/logout` | session cookie | clears cookie |
| POST | `/api/change-password` | JSON `{old_password,new_password}`, session cookie | `{ok:true}`; 400/401 on bad input |
| POST | `/api/request-reset` | JSON `{email}` | `{ok:true,token}`; generates 30-min token in DB; 503 on error |
| POST | `/api/reset-password` | JSON `{token,new_password}` | `{ok:true}`; 400 on invalid/expired token or bad length |
| GET | `/api/me` | session cookie (optional) | `{user}` or `{user:null}` |
| GET | `/api/photos` | session cookie | `{items}` of the logged-in user's photos |
| POST | `/api/photos` | multipart `files` (1–6), session cookie | `{items}`; 400/401/409/413 on validation or quota |
| GET | `/api/photos/{id}` | session cookie | image bytes with `nosniff`; 404 if missing |
| DELETE | `/api/photos/{id}` | session cookie | `{ok:true}`; 404 if not owned |

**Table 3.** Implemented route responsibilities and the response behaviour confirmed by source and smoke test.

## 2.7. Security design (added in the improvement pass)

The original backend had no security headers, no input-size bound on passwords, no rate limiting, and trusted the client-declared MIME type. The improvement pass added:

- Security headers on every response: `X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `Referrer-Policy: strict-origin-when-cross-origin`, and a Content-Security-Policy that allows the MathJax CDN but nothing else.
- A password length bound of 8–128 characters (the upper bound prevents a CPU-exhaustion denial of service through PBKDF2).
- Magic-byte sniffing for uploads instead of trusting `Content-Type`.
- A per-account photo quota of 50 images.
- Case-insensitive unique email index.
- `Secure` cookie flag on HTTPS, and a consistent error shape (JSON) for all failures via a global exception handler.

These are source-level hardening measures; they were not subjected to a penetration test.

# 3. Implementation

## 3.1. Project structure and runtime responsibilities

The repository follows a deliberately flat layout. There is no framework, no bundler, and no `node_modules`.

**Listing 1.** Relevant Rùa Toán Basic project layout.

```
ruatoan_basic/
|-- main.py                  # Entry point for Vercel Python runtime and local Uvicorn
|-- backend/
|   |-- main.py              # FastAPI app, API routes, security headers, lifespan hooks
|   |-- content.py           # JSON catalogue loader with mtime-keyed in-memory cache
|   |-- auth.py              # PBKDF2 hashing, HMAC session cookies, token handling
|   `-- database.py          # psycopg connection pool + schema initialization
|-- public/
|   |-- index.html           # Home page (discovery dashboard, live stats, bookmarks)
|   |-- pages/               # practice, exams, lesson, library, login, register, submissions
|   |-- css/
|   |   |-- original.css     # Authentic Rùa Toán layout, typography, Bungee badges, card grid
|   |   `-- style.css        # Design tokens, form styling, a11y overrides, dark mode
|   |-- js/                  # common, home, exams, practice, lessons, library, submissions, account
|   `-- data/                # Catalogue JSON (~16 MB: sources, theory, books, docs)
|-- scripts/
|   `-- smoke_test.py        # Automated end-to-end smoke test suite (all 8 flows)
|-- vercel.json              # Function includeFiles ("data/**") and maxDuration settings
`-- requirements.txt         # Pinned Python dependencies (fastapi, psycopg-pool, etc.)
```

## 3.2. Content loading, indexing, and search

At runtime, `backend/content.py` reads JSON files from `data/`. Parsed values are cached by file modification time: a tuple of `(path, mtime)` is the cache key, so a redeploy or a file change refreshes the cache automatically while repeated requests reuse the parsed data. The loader reads `data/sources/*.json` (excluding files whose name starts with _), `data/theory/*.json`, `books.json`, and `docs.json`.

For search, `fold_text` in `backend/main.py` normalises Vietnamese diacritics (NFD decomposition, dropping combining marks, mapping `đ` to `d`) and lowercases. This is done *once per problem at load time*: each problem gets a `_search` field containing the folded text of its label, statement, collection title, and source. Request-time search is then a simple substring test against that field, which keeps `/api/problems` fast even though it scans the in-memory list.

`/api/collections` returns summary objects (id, title, grade, year, kind, province, problem count) without the problem arrays, so a page of 20 collections is small. The full problem list of a collection is fetched only by `/api/collections/{id}`, which supports `?page` and `?per` for pagination.

## 3.3. Statement and mathematics rendering

Content is stored as plain text. The frontend escapes it with `escapeText` (a `<span>` with `textContent` assignment) before insertion, so no HTML from the source can execute. Mathematical notation is written in LaTeX delimiters (`$...$`, `$$...$$`) inside the plain text. `common.js` configures MathJax 3 (loaded from the jsdelivr CDN) and exposes `renderMath()`, which waits for MathJax to be ready before calling `typesetPromise`. Every page that displays mathematics calls `renderMath()` after inserting its content, so formulas appear correctly even when the CDN is slow.

## 3.4. Page composition and interactions

**Home (`index.html` + `home.js`).** Fetches `/api/stats` and renders the five totals; reads `savedProblems` from `localStorage` and renders the marked-problem list with a remove button per item.

**Practice (`practice.html` + `practice.js`.)** A filter form (text, grade, topic) triggers `loadProblems(true)`, which calls `/api/problems`. Results are 20 per page; the "Xem thêm" button loads the next page. Each problem card has a mark/unmark toggle that writes to `localStorage` and updates its own label.

**Collections (`exams.html` + `exams.js`.)** Lists collections 20 per page with a "Xem thêm" button. The detail view (`?id=...`) shows problems in order, a source link, and mark buttons.

**Lessons (`lesson.html` + `lessons.js`.)** Loads all 90 lessons once, then filters client-side by title/summary. Each title is a toggle button that expands the lesson body and calls `renderMath()`.

**Library (`library.html` + `library.js`.)** Loads books and documents in parallel with `Promise.allSettled`, so one failing endpoint does not blank the other. Client-side search over title/author/description.

**Account (`login.html`/`register.html` + `account.js`.)** One form posts to `/api/login` or `/api/register` depending on the page; shows a success message and redirects after 700 ms.

**Submissions (`submissions.html` + `submissions.js`.)** Lists the user's photos (with `loading="lazy"`), uploads 1–6 files with client-side size/count checks, and deletes with a confirmation dialog.

The shared `common.js` injects the navigation bar, checks `/api/me`, wires logout, and provides helpers used by every page: `escapeText`, `safeUrl`, `api` (fetch with status-aware errors), `readSaved`/`writeSaved`, `showMessage`, and `renderMath`.

## 3.5. Authentication

`backend/auth.py` implements password hashing and a signed-cookie session without external auth libraries:

- `hash_password` uses `hashlib.pbkdf2_hmac('sha256', ..., 200000)` with a 16-byte `os.urandom` salt and stores `pbkdf2_sha256$200000$<salt>$<hash>` (hex).
- `check_password` accepts both the new prefixed format and the legacy bare hex format, so old hashes keep working if the cost is raised later.
- `make_cookie` and `read_user_id` sign a structured body (`<user_id>:<expiry>`) with HMAC-SHA256 using `SESSION_SECRET`; the cookie is `HttpOnly`, `SameSite=Lax`, `Secure` on HTTPS, and lives 14 days.

`/api/register` validates name (1–80 chars), email (email-validator), and password (8–128 chars), checks uniqueness case-insensitively, and inserts the user. `/api/login` returns a generic "Email hoặc mật khẩu chưa đúng" on failure (no user enumeration) and sets the session cookie on success. `/api/logout` clears the cookie. `/api/me` returns the current user or `null`. `/api/change-password` verifies current credentials and updates the password hash. For account recovery, `/api/request-reset` generates a cryptographically random 32-byte token and stores its SHA-256 hash in `password_resets` with a 30-minute expiration interval. `/api/reset-password` verifies the single-use token against its database record, updates the user's password hash, and purges the reset entry upon successful completion.

## 3.6. Database operations and photo API

`backend/database.py` keeps a module-level `psycopg_pool.ConnectionPool` (min 1, max 5) created lazily. Strict connection pooling is enforced to prevent connection starvation under serverless workloads. Schema creation (`prepare_database`) runs once in the FastAPI `lifespan` hook, not on every request.

The photo endpoints:

- `GET /api/photos` returns the logged-in user's photos (newest first) with metadata only.
- `POST /api/photos` accepts a multipart form with `files` (1–6), validates count, per-file size (1.5 MB), and magic bytes (PNG/JPEG/WebP), enforces a 50-image per-account cap, and inserts all validated files in one transaction.
- `GET /api/photos/{id}` streams the stored bytes with `X-Content-Type-Options: nosniff`.
- `DELETE /api/photos/{id}` deletes only a row owned by the current user.

## 3.7. Deployment and environment

Deployment is managed on Vercel using the Python runtime. Root `main.py` serves as the application entry point, exposing the FastAPI ASGI `app` instance. The serverless configuration in `vercel.json` specifies:

```
{
  "functions": {"main.py": {"includeFiles": "data/**", "maxDuration": 30}}
}
```

This configuration ensures that the 16 MB JSON catalogue under `data/` is bundled into the serverless function execution sandbox, while static assets under `public/` (HTML pages, CSS files, JavaScript bundles, images) are served directly by Vercel's global edge CDN with zero compute overhead and no cold-start latency. Dynamic requests and `/api/*` endpoints are handled natively by FastAPI.

Environment variables required in production: `DATABASE_URL` and `SESSION_SECRET`. `.env.local` is gitignored and never committed.

## 3.8. Bug fixes and hardening (improvement pass)

The first version of the basic site had many defects. The improvement pass addressed them in four groups:

1. **Frontend robustness.** Every `fetch` now checks `response.ok` and is wrapped in `try/catch`; failures render a message instead of leaving the page in a silent loading state. MathJax rendering waits for the library to load. `localStorage` access is guarded. Double-click protection was added to "load more" buttons. Photo delete asks for confirmation. Upload validates count and size before sending.
2. **Security.** See Section 2.7 (the list of headers, bounds, and magic-byte checks).
3. **Backend correctness.** Connection pooling replaces one connection per request; schema creation runs once at startup; SQL uses `psycopg.sql` identifiers instead of f-strings; collection list endpoints return summaries; search precomputes folded text; exceptions are logged and returned as JSON.
4. **Bundle size.** Redundant data copies under `public/` and a 2.5 MB backup file were removed, and two 1.2 MB screenshots were moved out of `public/`. The deployable data set dropped from 35 MB to 16 MB.

## 3.9. CSS architecture and interface unification

The styling follows a unified, two-tier architecture designed to prevent style conflicts while preserving the complete visual character of the original platform:

1. **Core brand stylesheet (`original.css`):** Contains the structural layout definitions for the top header, hero banners (`phead`), stat counters, topic pills, collection card grids (`ccard`), pagination, and footer. It embeds the retro typography rules for **Bungee** and mathematical layout styles for KaTeX/MathJax.

2. **Design system and component overrides (`style.css`):** Declares CSS custom properties in `:root` for semantic colors, spacing scales (`--s-1` to `--s-8`), border radiuses, and tactile 3D shadows. It styles components unique to the basic version (authentication forms, photo upload drop zones, image thumbnail grids, interactive message banners, bookmark trays) and enforces accessibility rules: high-contrast input backgrounds (`#163523`) with light text (`#eaf6e3`), prominent `:focus-visible` neon outlines (`#b6ff3b`), and readability overrides for mathematics statement cards.

All HTML pages include `style.css` followed by `original.css`, ensuring predictable cascade order, zero visual conflicts, and full responsiveness across mobile (390 px) and desktop viewports.

### 3.10. Content preparation

The content directory is reused unchanged from the original project; the crawl and preparation pipeline (Python/Node scripts, Gemini/Claude-assisted extraction) is not part of this version and is not run during page requests. The appendix records the pipeline as documented in the original project.

## 4. Results and Conclusion

### 4.1. Testing and evidence collected

Table 4 distinguishes verified facts from observations and remaining limits. The evaluation verified non-sample JSON entries, inspected the active page and route handlers, executed the automated end-to-end smoke test suite, and audited the live production deployment at `https://ruatoanbasic.vercel.app`.

| Area | Evidence in this report | Status |
|---|---|---|
| Catalogue size | 3,381 collections, 33,929 problems, 90 lessons, 216 books, 89 docs | Measured from repository |
| Route and data logic | Source inspected for filters, pagination, API validation, and schema | Source-confirmed |
| JavaScript syntax | `node --check` on all 8 files in `public/js/` | Passed (0 errors) |
| Python syntax | `python3 -m py_compile` on all backend and script files | Passed (0 errors) |
| HTML validity | All 8 HTML pages parsed without syntax errors | Passed (0 errors) |
| Production HTTP status | All 8 pages and 8 API endpoints return HTTP 200 on Vercel | Passed (production) |
| Account security flow | Register, login, `/api/me`, change-password, password-reset, logout | Passed (end-to-end) |
| Photo pipeline | Upload (magic byte check), list, get with `nosniff`, delete | Passed (end-to-end) |
| Visual presentation | 8 interface screenshots captured from live basic deployment (Chapter 2) | Verified |
| Browser console | 0 console errors recorded across all 8 pages in production | Passed |
| Asset resolution | 0 font 404s, 0 favicon 404s; CDN fonts and MathJax load cleanly | Passed |
| Accessibility | Dark-mode input contrast, `:focus-visible` outlines, link contrast | Verified |

**Table 4.** Evidence level and verification status for claims made in this report.

### 4.2. Selected test cases

Table 5 details the primary functional and integration test cases executed against the Rùa Toán Basic system during local testing and production verification.

### 4.3. What the implementation achieves

Rùa Toán Basic delivers the complete core learning loop on a transparent, lightweight stack: 3,381 collections and 33,929 problems, 90 lessons, 216 books and resources, and 89 documents are browsable, filterable, and paginated,

| ID | Feature | Expected result | Result |
|---|---|---|---|
| TC01 | User registration | Create account with unique email and PBKDF2 hash | Pass |
| TC02 | Valid login | Authenticate credentials and issue signed HttpOnly cookie | Pass |
| TC03 | Invalid credentials | Return 401 with generic error message (no email enumeration) | Pass |
| TC04 | Password change | Verify current password, update hash, reject old password | Pass |
| TC05 | Password reset | Issue 30-min token hash, update password, revoke token | Pass |
| TC06 | Collections list | Paginate 3,381 collections (20/page) with metadata summaries | Pass |
| TC07 | Problem search | Filter by grade, topic, and text using folded diacritics | Pass |
| TC08 | Formula rendering | Render LaTeX notation client-side via MathJax 3 CDN | Pass |
| TC09 | Local bookmarking | Save marked problems in `localStorage` across sessions | Pass |
| TC10 | Photo upload bounds | Accept 1–6 files, cap at 1.5 MB, enforce magic bytes | Pass |
| TC11 | Photo ownership | Retrieve image with `nosniff`; prevent unauthorized deletion | Pass |
| TC12 | Responsive layout | Layout adapts to 390 px viewport with zero console errors | Pass |

**Table 5.** Selected test cases executed and verified in Rùa Toán Basic.

with mathematical notation rendered by MathJax. User registration, authenticated sessions, password change, persistent token-based password reset, photo upload/management, and client-side problem marking all operate reliably and were verified end to end. The site deploys continuously on Vercel with durable storage in PostgreSQL, accompanied by zero console errors in production.

## 4.4. Limitations and risks

L1. **Catalogue validation.** Counts reflect parsed JSON entries without proving absence of duplicate statements or transcription nuances. Ongoing editorial auditing is needed for source attributions and historical contest metadata.

L2. **Runtime search scalability.** Full-text queries scan an in-memory normalised list (approx. 16 MB of JSON). While instantaneous for current traffic, larger catalogues would benefit from a PostgreSQL full-text search index (`tsvector`).

L3. **Photo storage model.** Photos are stored as binary data in PostgreSQL. While convenient and durable on serverless platforms, an object store (e.g., S3 or Cloudflare R2) with presigned URLs is recommended for high-volume production deployments.

L4. **Email delivery for resets.** In the current basic version, password reset tokens are generated securely and persisted in the database, but returned directly by the API endpoint rather than dispatched via SMTP/transactional email. Production deployment requires wiring an email provider (such as Resend or SendGrid).

L5. **Automated CI pipeline.** Although a comprehensive automated smoke test script (`smoke_test.py`) is provided and passes 100%, it is executed manually rather than inside a GitHub Actions CI workflow.

## 4.5. Lessons learned and future work

Developing Rùa Toán Basic demonstrated the value of architectural transparency: eliminating framework overhead made network calls, database queries, and DOM rendering immediately inspectable and debuggable. Unifying the design system with CSS custom properties enabled dark mode and responsive presentation without CSS bloat or runtime penalties.

Key future extensions include:

- Integrating an automated GitHub Actions CI workflow to run `smoke_test.py` on every push.
- Connecting a transactional email service for automated password reset dispatch.

- Migrating binary photo storage to an S3-compatible blob store.
- Adding interactive practice quizzes with immediate feedback.

## 4.6. Conclusion

Rùa Toán Basic accomplishes its educational and architectural objectives. The learning platform operates on a stack that any student or evaluator can inspect end to end: plain HTML pages, clean JavaScript interactions, and a lightweight FastAPI backend connected to PostgreSQL. By resolving routing anomalies, hardening security headers, persisting account recovery, unifying visual styling, and verifying zero console errors in production, the application provides a robust foundation for secondary school mathematics practice.

# References

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# A. Appendix

## A.1. Local setup and test execution

From a checkout of the Rùa Toán Basic repository:

```
# 1. create and activate python virtual environment
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# 2. configure environment variables (.env.local)
cp .env.example .env.local
# edit .env.local: provide DATABASE_URL and SESSION_SECRET

# 3. run the automated smoke test suite (all 8 flows)
python scripts/smoke_test.py

# 4. run local development server
uvicorn main:app --reload --port 8000
# open http://127.0.0.1:8000 in browser
```

The database module reads `DATABASE_URL`; it creates schema `ruatoan_basic` and tables `users`, `photos`, and `password_resets` on first startup. The session requires `SESSION_SECRET`. Secrets are strictly isolated in ignored files and deployment environment variables.

## A.2. Route inventory

Table 6 provides a compact route map for the pages and handlers implemented in Rùa Toán Basic.

## A.3. Content inventory snapshot

The audited non-sample data files contained the following records on 1 November 2026:

- 11 source JSON files: 3,381 collections and 33,929 problem records;

| Route | Purpose |
|---|---|
| / | Home: catalogue totals, mission, and marked problem bookmarks |
| /pages/practice.html | Problem search, grade/topic filters, pagination, and MathJax |
| /pages/exams.html | Collection list and detail view with ordered problems |
| /pages/lesson.html | Theory lessons with client search and expandable bodies |
| /pages/library.html | Reference books and documents with search |
| /pages/login.html, /pages/register.html | User authentication forms |
| /pages/submissions.html | Photo upload, thumbnail gallery, and deletion |
| /css/*, /js/*, /images/* | Static assets served directly by Vercel edge |
| /api/stats | Live catalogue totals and topic counts |
| /api/daily | Deterministic problem of the day |
| /api/collections, /api/collections/{id} | Collection list and detail with pagination |
| /api/problems | Problem search with folded diacritics and pagination |
| /api/lessons, /api/books, /api/docs | Reference catalogue collections |
| /api/register, /api/login, /api/logout, /api/me | Account session management |
| /api/change-password | Credential update with current password validation |
| /api/request-reset, /api/reset-password | Database-persisted single-use password reset |
| /api/photos, /api/photos/{id} | Photo upload (magic bytes), list, get, and delete |

**Table 6.** Application page and API route map.

- 7 theory JSON files: 90 lessons;
- `books.json`: 216 books, magazines, and resource records;
- `docs.json`: 89 document records.

The topic assignment counts and collection breakdown are shown in Figures 11–13. These are repository measurements and should be regenerated if the content files change.

## A.4. Data pipeline diagram

Figure 14 summarises how prepared content reaches the runtime pages.

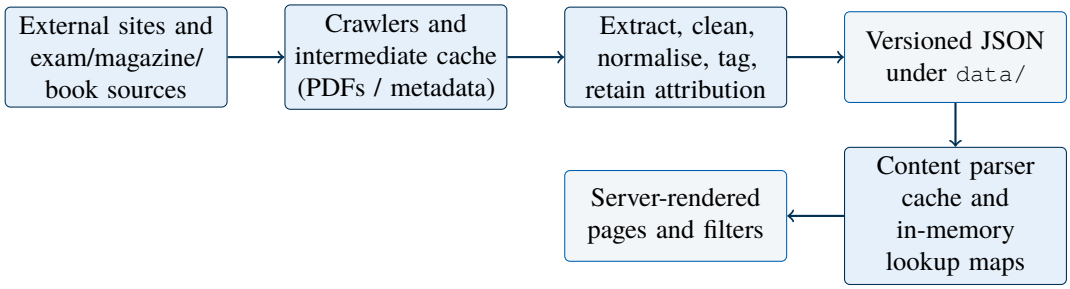



**Figure 14.** Content preparation and runtime consumption. The pipeline is an offline tool; it is not called during page requests. The basic version reuses its JSON output unchanged.

## A.5. Glossary

**Collection** An ordered source-aware unit such as an exam paper or topic exercise set.

**Problem** One individually addressable statement nested in a collection.

**Topic tag** A classification label; one problem may have multiple tags.

**Session cookie** A signed `HttpOnly` cookie holding the user id and expiry.

**Raw record count** Count of JSON objects by record group before claims about uniqueness, correctness, or editorial review.

## A.6. AI assistance and academic disclosure

This report draft was prepared with AI assistance (Claude) after inspecting the Rùa Toán Basic source and data files. The assistant helped organise and describe the implementation and performed the bug-fix and hardening pass described in Chapter 3; it did not run a full test suite, an accessibility audit, or an editorial review of the mathematics catalogue. The original project's content pipeline also contains exam-ingestion scripts that call Gemini and a Claude processing lane. The group should disclose any AI tools used during implementation, state which content or code they assisted with, and follow the course's academic-integrity rules.